



MMSQL: Evaluating and Enhancing LLMs for Multi-turn Text-to-SQL with Multiple Question Types

Ziming Guo, Chao Ma, Yinggang Sun, Tiancheng Zhao, Guangyao Wang, Hai Huang

Harbin University of Science and Technology, Harbin Institute of Technology



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Motivation

The Need for Robust Text-to-SQL Systems

Database schema given

Airlines

uid:airline id type:number PRIMARY KEY

Airline:airline name type:text

Abbreviation:abbreviation type:text

Country:country type:text

Airports

City:city type:text

AirportCode:airport code type:text PRIMARY KEY

AirportName:airport name type:text

Country:country type:text

CountryAbbrev:country abbrev type:text

Flights

Airline:airline type:number PRIMARY KEY

FlightNo:flight number type:number

SourceAirport:source airport type:text

DestAirport:destination airport type:text

Foreign keys:

flights.'DestAirport' = airports.'AirportCode'

flights.'SourceAirport' = airports.'AirportCode'

Turn1: Unanswerable



How many flights take off **from** **airport APG** each day?



SELECT date, COUNT(*) AS flight_count
FROM flights WHERE departure_airport =
'APG' GROUP BY date



I'm sorry, the database does not include
information about flight schedules or
dates.

Turn2: Answerable



How many airlines have flights
departing from **here**?

SELECT COUNT(T1.Airline) FROM
Airlines AS T1 JOIN Flight AS T2
ON T1.uid = T2.Airline WHERE
T2.sourceAirport like "%APG%"



Turn3: Ambiguous



What is the **flight number of Delta Airlines**?



SELECT T2.FlightNo FROM airlines AS T1
JOIN flights AS T2 ON T1.uid = T2.Airline
WHERE T1.Airline = 'Delta Airlines'



Your question is ambiguous. Are you asking
for **all flight numbers operated by Delta Airlines**, or specifically **those departing from APG**? If you're asking for the former, the SQL query would be [SQL1]. If you're asking for the latter, the SQL query would be [SQL2].

Turn4: Improper



Thanks!

You're welcome! If you have any
more questions, feel free to ask.



Motivation

The Need for Robust Text-to-SQL Systems

- ▶ Progress in Text-to-SQL Systems: Empowering non-technical users to query databases using natural language.
- ▶ Challenges: Handling **dynamic and uncertain real-world user queries**.
- ▶ Multi-turn conversations
- ▶ Ambiguous or unanswerable questions
- ▶ Current models' limitations leading to unreliable predictions
- ▶ Objective: Develop more robust and dependable text-to-SQL systems.

Dataset

Introduction to MMSQL

- ▶ Question Types: **Answerable / Unanswerable / Improper / Ambiguous**
- ▶ Dataset Generation: QDA-SQL to generate multi-turn Q&A pairs.
- ▶ Original data: Transformed samples from SPARC and CoSQL.
- ▶ Dataset Characteristics: Complex multi-turn and multi-type scenarios.
- ▶ Download (dataset and evaluation scripts):



Github:

- mcxiaoxiao.github.io/MMSQL



Hugginface:

- `from datasets import load_dataset`
- `ds = load_dataset("GuoZiming/MMSQL")`

	SParC	CoSQL	NoisySP	AmbiQT	AMBROSIA	MMSQL
# Dialogues	4,298	3,007	-	-	-	6493
Total # turns	12,726	15,433	15,598	3,046	4,242	38,666
Avg. # Q turns	3.0	5.2	1	1	1	6.0
Avg. Q len	10.2	11.2	-	-	-	11.4
Ans. Q type	✓	✓	✗	✗	✓	✓
Amb. Q type	✗	✓	✓	✓	✓	✓
Una. Q type	✗	✓	✓	✗	✗	✓
Imp. Q type	✗	✓	✗	✗	✗	✓



MMSQL Test Set (149)



MMSQL Train Set (6493)

Evaluation Metrics

Metrics Overview

- ▶ **Exact Matching (EM)**: All components of the predicted SQL must match the reference query (excluding values).
- ▶ **Execution Accuracy (EX)**: Proportion of SQL queries with identical execution results.
- ▶ **Interaction Execution Accuracy (IEX)**: All SQL queries in a multi-turn interaction execute correctly.
- ▶ **Dual Assessment of Question Type Detection and Execution Accuracy (TDEX)**: Evaluates both question type classification and execution accuracy.
- ▶ **Response Quality Score (RQS)**: Measures the quality of natural language responses using an LLM-assisted rating method.

$$\text{TDEX} = \frac{1}{N} \sum_{i=1}^N \begin{cases} \varepsilon_{\text{exec}}(S_i, \hat{S}_i) & \text{(a)} \\ \delta_{\text{type}}(C_i, \hat{C}_i) & \text{(b)} \end{cases}$$

- (a) $C_i = \text{'Answerable' or 'Ambiguous'}$
(b) otherwise

Model Performance

Key Insights

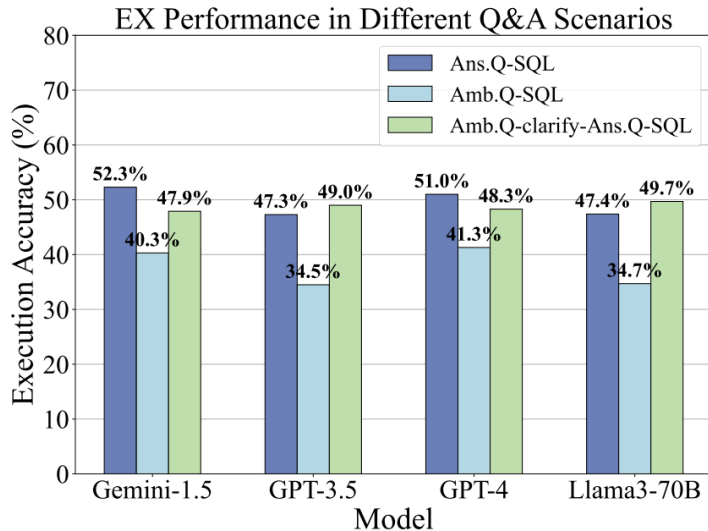
- ▶ Closed-source models (e.g., GPT-4 Turbo) have a slight performance edge.
- ▶ Open-source models (e.g., Llama3-70B) are rapidly catching up.
- ▶ All models struggle with unanswerable and ambiguous questions.

Model	Ans.		Una.		Amb.		Imp.		F1
	Prec	Rec	Prec	Rec	Prec	Rec	Prec	Rec	
GPT-4 Turbo	<u>90.3</u>	89.8	25.9	<u>70.0</u>	56.9	38.4	100.0	<u>98.0</u>	68.2
GPT-3.5 Turbo	88.8	89.2	16.0	65.0	64.3	20.9	100.0	96.0	61.1
Gemini-1.5 Flash	85.9	<u>95.8</u>	26.3	50.0	58.3	8.1	100.0	95.4	59.3
Llama3-70B	84.9	95.2	<u>27.5</u>	55.0	80.0	9.3	100.0	92.7	59.8
Llama3-8B	88.3	93.2	21.4	60.0	83.3	23.3	100.0	96.7	<u>64.2</u>
SQLCoder-8B	84.6	99.6	77.8	35.0	0.0	0.0	100.0	<u>98.0</u>	59.7
Codellama-7B	93.9	16.5	4.3	85.0	<u>96.6</u>	66.3	56.8	100.0	46.9
Mistral-7B-v0.2	82.1	57.6	4.7	55.0	100.0	<u>50.0</u>	79.1	77.5	55.3

Model Performance

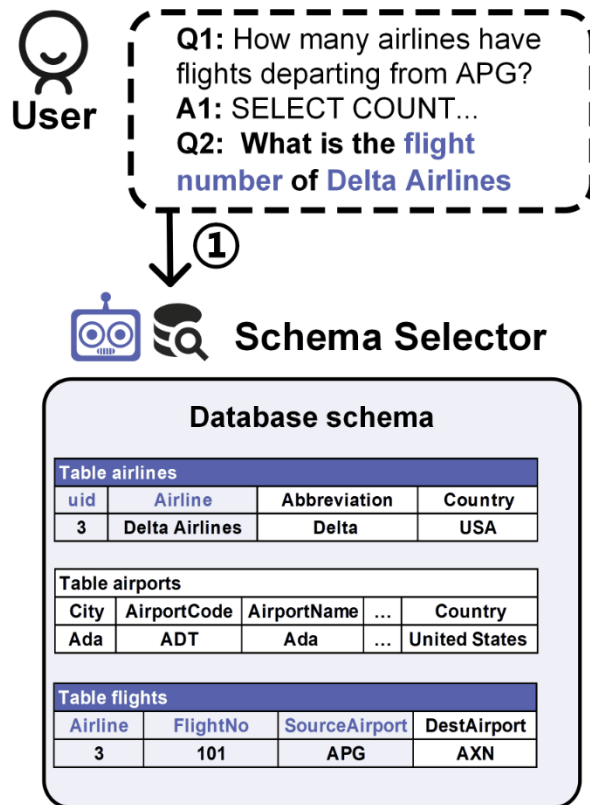
Key Insights

- ▶ Closed-source models (e.g., GPT-4 Turbo) have a slight performance edge.
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- ▶ All models struggle with unanswerable and ambiguous questions.
- ▶ **Clarification processes in multi-turn interactions improve query precision.**



Multi-Agent Framework

Framework Overview



Select relevant tables and fields

Conclusion

Summary of Findings

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Ziming Guo[✉], Chao Ma*, Yinggang Sun[†], Tiancheng Zhao*, Guangyao Wang[✉], Hai Huang*

*School of Computer Science, Harbin University of Science and Technology

Harbin, China 150040

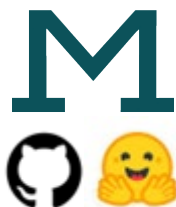
Email: 2204050108@stu.hrbust.edu.cn, machao8396@163.com,
1783467143@qq.com, 2204050124@stu.hrbust.edu.cn, hust_hh@vip.163.com

[†] Faculty of Computing, Harbin Institute of Technology, Harbin, China 150001

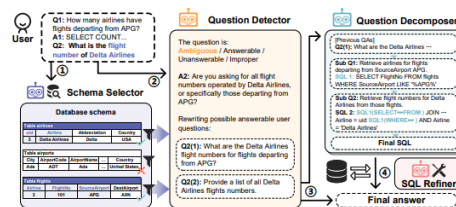
Email: 23b903085@stu.hit.edu.cn



Real-world challenges&insight



Datasets



Multi-Agent Framework

Q&A

Engage with Us

- ▶ **Any questions?** orlosziming@outlook.com
- ▶ **Want to contribute?** Submit issues or PRs to our repo



`github.com/mcxiaoxiao/MMSQL`